

Yuhang Zhang

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EDUCATION

Texas A&M University

College Station, TX

Ph.D. Candidate, Mechanical Engineering; GPA: 4.0/4.0

Jan 2023 – Feb 2027 (Expected)

- Dissertation: *Dynamic Modeling and Optimal Control of District Heating and Cooling Systems (working title)*
- Advisor: Prof. Zheng O'Neill

Tongji University

Shanghai, China

Master of Engineering in Power Engineering; GPA: 4.9/5.0

Sep 2019 – Mar 2022

- Thesis: *Study on Leakage Diagnosis Method for Air Conditioning Water System Network*
- Advisor: Prof. Zhengwei Li

Tongji University

Shanghai, China

Bachelor of Science in Building Environment and Energy Engineering; GPA: 4.6/5.0

Sep 2015 – Jul 2019

RESEARCH INTERESTS

- District energy, heat pumps, and thermal storage
- Predictive control of thermal systems
- Building decarbonization and waste-heat recovery

RESEARCH EXPERIENCE

NSF PIRE: Building Decarbonization via AI-empowered District Heat Pump Systems

National Science Foundation (NSF)

Jan 2023 – Present

- Built the Modelica district heat-pump virtual testbed and led analysis of network designs and operating strategies.
- Developed and evaluated model predictive control for district-loop temperature management, reducing modeled total system energy use by up to 10% against baseline control.
- Modeled geothermal and data-center waste-heat integration; generated fault datasets used by FDD collaborators.

DoD ESTCP: Building Energy Efficiency through Thermal Microgrids – Fort Hood Phase I Feasibility Study

Department of Defense (DoD) ESTCP

Dec 2023 – Present

- Built Modelica virtual testbeds for 30+ Fort Hood buildings and reconstructed hourly loads with XGBoost.
- Developed simulated model predictive control with an LSTM-based model for district-loop operation.
- Compared retrofits; one building cluster showed nearly 68% lower modeled annual district thermal-system energy input than its existing-system baseline.
- Supported NIST BLCC life-cycle cost comparisons across alternatives with different study periods.

DOE: Solar-Geothermal District Heating and Cooling for Citizen Potawatomi Nation

Department of Energy (DOE)

Dec 2023 – Nov 2024

- Developed Modelica models of a proposed single-pipe geothermal district network, integrating borefields, building heat pumps, and the distribution loop.
- Analyzed annual borefield thermal balance and compared pumping and loop-temperature control strategies in simulation.
- Developed photovoltaic-thermal collector models and integrated them into a separate 5GDHC virtual testbed documented in a 2025 conference paper.
- Phase I simulations projected about 38% lower electricity use than existing heating, cooling, and domestic-hot-water systems for the proposed all-electric design.

Master's Thesis: Study on Leakage Diagnosis Method for Air Conditioning Water System Network

Tongji University

Sep 2020 – Mar 2022

- Calibrated a Modelica hydraulic-network model against measured normal-operation data from a metro-station chilled-water system.
- Developed a two-stage neural-network method to locate simulated pipe leaks, achieving 86.96% average pipe-identification accuracy.
- Evaluated the effects of sensor count and accuracy, training-set size, and data noise on diagnostic reliability.

Public Building Energy Audits – Tongji University Project for Shanghai Dongfang Yanhua

Shanghai, China

Jul 2019 – Dec 2019

- Co-led site surveys and energy audits of 22 Shanghai public buildings through a Tongji University project for Shanghai Dongfang Yanhua.
- Prepared audit reports and evaluated HVAC/lighting upgrades with projected energy and cost savings.

PROFESSIONAL EXPERIENCE

PhD Intern

Jun 2025 – Aug 2025

Pacific Northwest National Laboratory (PNNL)

Richland, WA

- Built a Python-based workflow for DOE's thermal-energy-storage sizing tool, generating 240 case configurations to compare storage capacities and time-of-use pricing.
- Developed chilled-water thermal energy storage controls and compared modeled daily utility costs against schedule-based control across four representative-day simulations.

Graduate Research Assistant

Jan 2023 – Present

Texas A&M University

College Station, TX

- Develop Modelica virtual testbeds and Python workflows across NSF-, DoD-, and DOE-funded district-energy projects.
- Developed open-source water-to-air heat-pump models validated against manufacturer and physical measurements; capacity CVMSE was 3.8% in cooling and 8.2% in heating.
- Combine operational data, simulation, and predictive control to evaluate retrofit designs and operating strategies.

Smart Building Engineer

Jul 2022 – Dec 2022

Envision Group

Shanghai, China

- Led product development of HVAC optimization and FDD features, including AHU and chiller control, for an IoT smart-building platform.
- Translated client needs into requirements, defined control logic and interface prototypes, and coordinated algorithm validation and development handoff.

PUBLICATIONS

Journal Articles

- [1] Liu, M., **Zhang, Y.**, Yang, Z., & O'Neill, Z. (2026). Comparative study of energy flexibility in model predictive control of single- and variable-speed heat pumps with active thermal energy storage. *Energy*, 358, 141387. DOI
- [2] Liu, M., **Zhang, Y.**, Yang, Z., & O'Neill, Z. (2026). A nationwide assessment of energy flexibility from predictive-controlled heat pump and active thermal energy storage system for building electrification. *Building Simulation*, 19(4), 1131–1154. DOI
- [3] **Zhang, Y.**, Liu, M., O'Neill, Z., & Wen, J. (2024). Temperature control strategies for fifth generation district heating and cooling systems: A review and case study. *Applied Energy*, 376, 124156. DOI
- [4] Liu, R., **Zhang, Y.**, & Li, Z. (2022). Leakage diagnosis of air conditioning water system networks based on an improved BP neural network algorithm. *Buildings*, 12(5), 610. DOI

Conference Proceedings

- [1] Jiang, Z., Song, R., Li, G., **Zhang, Y.**, O'Neill, Z., Wang, X., Goldfeder, J., & Dong, B. (2026). BESTOpt: A Modular, Physics-Informed Machine Learning based Building Modeling, Control and Optimization Framework. *SimBuild Conference 2026 (IBPSA-USA)*, 12, 786–797. DOI
- [2] Guan, H., **Zhang, Y.**, O'Neill, Z., & Dong, B. (2026). A Scientific Machine Learning Approach for Geothermal Borefield Thermal Modeling via Residual Learning. *Proceedings of the 9th International High Performance Buildings Conference at Purdue*, Paper 3273, 1–10.
- [3] **Zhang, Y.**, Liu, M., O'Neill, Z., & Wen, J. (2025). Operating temperature optimization for a 5th-generation district heating and cooling network using model predictive control. *2025 ASHRAE Winter Conference*, Orlando, FL. DOI
- [4] Li, G., **Zhang, Y.**, Liu, M., O'Neill, Z., Song, L., Wang, G., Cai, J., & Wu, X. (2025). Modeling and integration of photovoltaic thermal (PVT) hybrid solar collectors in a fifth-generation district heating and cooling network (5GDHC). *Proceedings of Building Simulation 2025*. DOI
- [5] **Zhang, Y.**, Liu, M., Yang, Z., Calfa, C., & O'Neill, Z. (2024). Development and validation of a water-to-air heat pump model using Modelica. *Proceedings of the American Modelica Conference 2024*, 119–126. **Best Student Paper**. DOI
- [6] **Zhang, Y.**, Liu, M., O'Neill, Z., & Wen, J. (2024). A comparative analysis of energy performance of ultra-low temperature district heating and cooling networks with different substation configurations. *ASHRAE Transactions*, 130, 796–805. DOI

Under Review

- [1] Guan, H., **Zhang, Y.**, O'Neill, Z., & Dong, B. (2026). Adaptive Fusion: A Scientific Machine Learning Framework for Geothermal Borefield Thermal Modeling. (*Under review at Energy*).
- [2] **Zhang, Y.**, Liu, M., O'Neill, Z., & Wen, J. (2026). A physics-based virtual testbed for fifth-generation district heating and cooling systems: a model predictive control case study of network temperature optimization. (*Under review at Building Simulation*).
- [3] **Zhang, Y.**, Guan, H., O'Neill, Z., & Wen, J. (2026). A unified physics-based framework for system-level fault impact analysis in fifth-generation district heating and cooling systems. (*Under review at Energy & Buildings*).

CONFERENCE PRESENTATIONS (WITHOUT PROCEEDINGS)

- “Feasibility of Geothermal Borefield Integration in a 5th-Generation District Thermal Energy Network: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Geothermal Research Opportunities Symposium*, Apr 2025. **2nd Place, Poster Competition.**
- “Building Decarbonization with a 5th Generation District Heating and Cooling (5GDHC) System: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Conference on Energy, College Station, TX*, Sep 2024.
- “A Comparative Analysis of Energy Performance of Ultra-low Temperature District Heating and Cooling Networks with Different Substation Configurations” — *BAS2024 – Building Automation Symposium, Syracuse, NY*, May 2024.

ACADEMIC SERVICE

Peer Review

- Reviewer, *Science and Technology for the Built Environment* (2025–Present)
- Reviewer, ASHRAE Conferences (Winter & Annual, 2025–2026)
- Reviewer, AEI International Research & Design Conference (2026)
- Reviewer, *Journal of Engineering for Sustainable Buildings and Cities* (ASME) (2026)

ASHRAE Technical Committees

- Corresponding Member, TC 7.5 – Smart Building Systems (2025–Present)
- Provisional Corresponding Member, TC 6.2 – District Energy (2023–Present)
- Provisional Corresponding Member, TC 6.9 – Thermal Storage (2023–Present)
- Provisional Corresponding Member, TC 6.8 – Geothermal Heat Pump Systems (2023–Present)
- Provisional Corresponding Member, TC 9.9 – Mission Critical Facilities, Data Centers, Technology Spaces and Electronic Equipment (2026–Present)

Leadership

- ASHRAE Student Branch Officer, Texas A&M University (2023–2024): organized monthly technical seminars and promoted student engagement in ASHRAE activities and conferences.

HONORS & AWARDS

- Pioneer Natural Resources Fellowship (2026–2027)
- ASHRAE Graduate Student Grant-in-Aid Award (2026–2027)
- J. George H. Thompson Graduate Fellowship (2025)
- Mechanical Engineering Graduate Summer Research Grant (2024–2025, two awards)
- Mechanical Engineering Graduate Student Travel Award (2023–2025, three awards)
- 2nd Place Poster Competition, Texas A&M Geothermal Research Symposium (2025)
- Best Student Paper, American Modelica 2024 Conference (2024)
- Tongji University Scholarships of Excellence (2015–2018, four awards)
- National Encouragement Scholarship (2015–2016)

TECHNICAL METHODS & TOOLS

Programming: Python, Git, MATLAB

Simulation: Modelica/Dymola, FMI/FMU co-simulation, EnergyPlus/OpenStudio

Methods: Model Predictive Control (MPC), Machine learning for thermal systems, Numerical optimization